

Class 12 NEB Computer Science Notes

1.2 Field, Record, Object, Primary Key, Candidate Key, Alternate Key (Detailed Notes)

1.2 FIELD, RECORD, OBJECT, PRIMARY KEY, CANDIDATE KEY, ALTERNATE KEY

1. FIELD (फिल्ड)

- A field is the smallest unit of data in a table.
- It represents a single attribute or property of an entity.

फिल्ड भनेको तालिकाको सबैभन्दा सानो इकाई हो। यो कुनै एक गुण वा विशेषतालाई जनाउँछ।

Example:
In Student table, Name, RollNo, Age, Class are fields.

RollNo	Name	Age	Class	Address
101	Ram	17	12	Kathmandu

Fields

2. RECORD (रेकर्ड)

- A record is a collection of related fields that describes a complete entity.

रेकर्ड भनेको सम्बन्धित फिल्डहरूको समूह हो, जसले एक सम्पूर्ण इकाई वा वस्तुको विवरण दिन्छ।

Example:
Each row in the above table is a record.

RollNo	Name	Age	Class	Address
101	Ram	17	12	Kathmandu
102	Sita	16	12	Lalitpur
103	Hari	17	12	Bhaktapur

Record

3. OBJECT (अब्जेक्ट)

- An object is a real-world entity about which data is stored.

अब्जेक्ट भनेको वास्तविक संसारको कुनै वस्तु वा इकाई हो, जसको बारेमा डाटा संग्रह गरिन्छ।

Example:
Student, Book, Teacher, Employee, Car etc. are objects.

Examples of Objects



4. KEYS IN DATABASE

4.1 PRIMARY KEY (प्राथमिक कुञ्जी)

- A primary key is a field (or combination of fields) that uniquely identifies each record in a table.
- It must contain unique values and cannot contain NULL.

प्राथमिक कुञ्जी भनेको त्यस्तो फिल्ड (वा फिल्डहरूको संयोजन) हो, जसले प्रत्येक रेकर्डलाई अद्वितीय रूपमा पहिचान गर्छ। यसमा अद्वितीय मान हुन्छ र NULL हुन सक्दैन।

RollNo (PK)	Name	Age	Class
101	Ram	17	12
102	Sita	16	12
103	Hari	17	12

4.2 CANDIDATE KEY (उम्मेदवार कुञ्जी)

- Candidate key is a field (or set of fields) that can uniquely identify a record.
- There can be one or more candidate keys in a table.

उम्मेदवार कुञ्जी भनेको त्यस्तो फिल्ड (वा फिल्डहरूको सेट) हो, जसले रेकर्डलाई अद्वितीय रूपमा पहिचान गर्न सक्छ। तालिकामा एक वा बढी उम्मेदवार कुञ्जी हुन सक्छन्।

RollNo	Email	Name	Age
101	ram@gmail.com	Ram	17
102	sita@gmail.com	Sita	16
103	hari@gmail.com	Hari	17

Here, RollNo and Email can be candidate keys.

4.3 ALTERNATE KEY (वैकल्पिक कुञ्जी)

- Alternate key is the candidate key that is not selected as the primary key.
- There can be one or more alternate keys.

वैकल्पिक कुञ्जी भनेको त्यस्तो उम्मेदवार कुञ्जी हो, जसलाई प्राथमिक कुञ्जीको रूपमा चयन गरिएको भएन। एक वा बढी वैकल्पिक कुञ्जी हुन सक्छन्।

RollNo	Email	Name	Age
101	ram@gmail.com	Ram	17
102	sita@gmail.com	Sita	16
103	hari@gmail.com	Hari	17

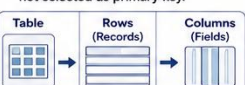
If RollNo is chosen as Primary Key, then Email becomes Alternate Key.

DIFFERENCES AT A GLANCE

Term	Meaning	Example	Key Points
Field (फिल्ड)	Smallest unit of data in a table (Represents an attribute)	Name, Age, Address, RollNo	- Single value - Column in a table
Record (रेकर्ड)	Collection of related fields that describes an entity	All details in a single row of a table	- Many fields combined - Row in a table
Object (अब्जेक्ट)	Real-world entity about which data is stored	Student, Book, Teacher, Employee	- Physical or conceptual thing - Described by records
Primary Key (प्राथमिक कुञ्जी)	Field that uniquely identifies each record	RollNo	- Unique - Cannot be NULL - Only one PK in a table
Candidate Key (उम्मेदवार कुञ्जी)	Field (or set of fields) that can uniquely identify a record	RollNo, Email	- One or more possible keys - Can be chosen as PK
Alternate Key (वैकल्पिक कुञ्जी)	Candidate key(s) not chosen as primary key	Email (if RollNo is Primary Key)	- Remaining candidate keys - Can be used if PK changes

KEY POINTS TO REMEMBER

- A Table is made of Columns (Fields) and Rows (Records).
- A Record represents one instance of an Object.
- Primary Key uniquely identifies each record and has unique values.
- Candidate Key can be more than one.
- Alternate Key is the candidate key not selected as primary key.



EXAMPLE TO UNDERSTAND KEYS

Suppose we have a table EMPLOYEE.

EmpID	Email	Mobile	Name
1	ram@gmail.com	9800000001	Ram
2	sita@gmail.com	9800000002	Sita
3	hari@gmail.com	9800000003	Hari

↓

Primary Key (can uniquely identify each employee) Candidate Key (can uniquely identify each employee) Candidate Key (can uniquely identify each employee) Not a Key (Names can be same)

If EmpID is chosen as Primary Key, then Email and Mobile are Alternate Keys.

**NEB TIP**

Learn definitions clearly and remember the differences. Diagram and examples are important for scoring good marks in the exam. Practice identifying fields, records, objects and different types of keys in various tables.

**IMPORTANT FOR EXAM**

Very Short, Short & Long Questions are frequently asked from these topics.

1. Field (फिल्ड)

Definition (English)

A **Field** is the smallest unit of data in a database table. It represents a single attribute (property or characteristic) of an entity.

Definition (Nepali)

फिल्ड भनेको डाटाबेसको तालिकामा रहेको सबैभन्दा सानो डेटा इकाइ हो। यसले कुनै वस्तु (Entity) को एउटा विशेषता (Attribute) जनाउँछ।

Example

Student Table

Roll No	Name	Age	Class	Address
101	Ram	17	12	Kathmandu

Here,

- Roll No → Field
- Name → Field
- Age → Field
- Class → Field
- Address → Field

Every **column** in a table is called a **Field**.

Characteristics of Field

- Represents one attribute only.
- Stores one type of data.
- Forms the columns of a table.
- Smallest logical unit of data.

Real-life Example

Student Admission Form

- Name
- Address
- Phone Number
- Date of Birth

Each item is a **Field**.

2. Record (रेकर्ड)

Definition (English)

A **Record** is a collection of related fields that contains complete information about one entity.

नेपाली परिभाषा

रेकर्ड भनेको सम्बन्धित फिल्डहरूको समूह हो, जसले एउटा वस्तु (Entity) सम्बन्धी सम्पूर्ण जानकारी दिन्छ।

Example

Roll No	Name	Age	Class
101	Ram	17	12

The entire row is one **Record**.

Another Example

Roll	Name	Address
1	Ram	Pokhara
2	Sita	Kathmandu

There are **2 Records**.

Characteristics

- Represents one complete object.
 - Made up of multiple fields.
 - Stored as a row.
-

Real-life Example

Employee Record

ID	Name	Salary
101	Hari	Rs.50,000

This whole row is one employee record.

Difference Between Field and Record

Field	Record
Column	Row
Smallest data unit	Collection of fields
One attribute	Complete information

Example: Name

Example: Ram,17,12

3. Object (अब्जेक्ट)

Definition (English)

An **Object** is a real-world entity about which information is stored in the database.

नेपाली परिभाषा

Object भनेको वास्तविक संसारको कुनै व्यक्ति, स्थान, वस्तु वा अवधारणा हो, जसको बारेमा डाटाबेसमा जानकारी राखिन्छ।

Examples

- Student
 - Teacher
 - Book
 - Employee
 - Hospital
 - Patient
 - Car
 - Product
-

Example

Suppose the object is **Student**.

Its fields are:

- Roll No
- Name
- Age
- Address
- Phone

These fields together describe the object "Student".

Real-life Example

Library Database

Object = Book

Fields

- Book ID
 - Book Name
 - Author
 - Price
-

4. Key (Key in Database)

A **Key** is one or more fields used to identify records in a table.

Keys help

- Find records
 - Update records
 - Delete records
 - Maintain relationships between tables
-

5. Primary Key (प्राथमिक कुञ्जी)

Definition

A **Primary Key** is a field (or combination of fields) that uniquely identifies each record in a table.

नेपाली परिभाषा

प्राथमिक कुञ्जी भनेको त्यस्तो फिल्ड हो जसले तालिकाको प्रत्येक रेकर्डलाई अद्वितीय रूपमा चिनाउँछ।

Characteristics

- Must be Unique
 - Cannot be NULL
 - Only one Primary Key per table
 - Used to identify records
-

Example

Roll No	Name	Age
101	Ram	17
102	Sita	18
103	Hari	17

Primary Key = **Roll No**

Because

- No duplicate
- No empty value

Why Name cannot be Primary Key?

Roll No	Name
101	Ram
102	Ram

Duplicate names are possible.

Therefore,

Name cannot uniquely identify students.

Advantages

- Prevents duplicate records
 - Easy searching
 - Faster retrieval
 - Maintains integrity
-

6. Candidate Key (उम्मेदवार कुञ्जी)

Definition

A **Candidate Key** is any field (or combination of fields) that can uniquely identify a record.

There may be **multiple candidate keys**.

नेपाली परिभाषा

उम्मेदवार कुञ्जी भनेको त्यस्तो फिल्ड हो जसले प्रत्येक रेकर्डलाई अद्वितीय रूपमा चिनाउन सक्छ।

Example

Student Table

Roll No	Email	Name
101	ram@gmail.com	Ram
102	sita@gmail.com	Sita
103	hari@gmail.com	Hari

Possible Candidate Keys

- Roll No
- Email

Both are unique.

Characteristics

- Unique
 - No NULL value
 - One of them becomes Primary Key
-

7. Alternate Key (वैकल्पिक कुञ्जी)

Definition

An **Alternate Key** is the candidate key that is **not selected** as the primary key.

नेपाली परिभाषा

वैकल्पिक कुञ्जी भनेको प्राथमिक कुञ्जीको रूपमा चयन नगरिएको उम्मेदवार कुञ्जी हो।

Example

Suppose

Candidate Keys

- Roll No
- Email

If Roll No becomes Primary Key,

then

Email becomes Alternate Key.

Diagram

Candidate Keys

Roll No

Email

Choose Roll No

↓

Primary Key = Roll No

↓

Alternate Key = Email

Difference Between Primary Key and Candidate Key

Primary Key

Candidate Key

Selected from candidate keys	Can become primary key
Only one	Can be many
Cannot be NULL	Cannot be NULL
Must be unique	Must be unique

Difference Between Candidate Key and Alternate Key

Candidate Key	Alternate Key
Can uniquely identify records	Candidate key not chosen as PK
One or more	One or more
One becomes PK	Remaining become Alternate Keys

Complete Example

Employee Table

EmpID	Email	Mobile	Name
101	ram@gmail.com	98000000001	Ram

102	sita@gmail.com	98000000002	Sita
103	hari@gmail.com	98000000003	Hari

Fields

- EmpID
- Email
- Mobile
- Name

Records

There are **3 Records**.

Object

Employee

Candidate Keys

- EmpID
- Email
- Mobile

Primary Key

EmpID

Alternate Keys

- Email
 - Mobile
-

Comparison Table

Term	Meaning	Example
Field	Single attribute (column)	Name
Record	Collection of fields (row)	Ram, 17, 12
Object	Real-world entity	Student
Primary Key	Unique identifier	Roll No
Candidate Key	Possible primary key	Roll No, Email
Alternate Key	Candidate key not selected as PK	Email

Advantages of Using Keys

- Uniquely identify records.
 - Prevent duplicate data.
 - Improve searching speed.
 - Maintain data integrity.
 - Establish relationships between tables.
 - Simplify updating and deleting records.
-

Frequently Asked NEB Exam Questions

2 Marks

1. Define Field with an example.
2. Define Record with an example.
3. What is an Object in DBMS?
4. Define Primary Key.
5. Define Candidate Key.
6. What is an Alternate Key?
7. Differentiate between Field and Record.
8. What are the characteristics of a Primary Key?

5 Marks

1. Explain Field, Record, and Object with suitable examples.
2. Explain Primary Key, Candidate Key, and Alternate Key with examples.
3. Differentiate between Primary Key and Candidate Key.
4. Explain the role of keys in a database.
5. Draw a table and identify the fields, records, object, primary key, candidate key, and alternate key.

Quick Revision

- **Field** → Column of a table; stores one attribute.

- **Record** → Row of a table; stores complete information about one entity.
- **Object** → Real-world entity (e.g., Student, Employee, Book).
- **Primary Key** → Unique field selected to identify each record; cannot be NULL or duplicate.
- **Candidate Key** → Any field(s) that can uniquely identify a record; there can be multiple.
- **Alternate Key** → Candidate key(s) that are **not** chosen as the Primary Key.